

# SHRAVANKUMAR NAGARAJAN

shravankumar.nagarajan@gmail.com | San Jose, CA, USA | LinkedIn | GitHub

## PROFESSIONAL SUMMARY

---

Software Engineer with 9+ years of experience designing and delivering scalable, production-grade systems across banking, publishing, and technology domains. Proven expertise in problem solving, system design, and distributed systems, with deep experience architecting high-performance, fault-tolerant applications that operate reliably at scale. Demonstrated building modular, reusable components and implementing real-time streaming architectures that deliver maintainability, scalability, and consistent performance in production environments.

## SKILLS

---

- **Programming Languages:** Java, Python, JavaScript, SQL, Groovy
- **Frameworks & Libraries:** Spring Boot, Project Reactor, FastAPI, React.js, Express.js, Apache Spark, REST APIs, GraphQL
- **Data & ML Tools:** Scikit-learn, Pytorch, Jupyter, Pandas, NumPy, LangChain, Power BI, Seaborn, Trino
- **DevOps & CI/CD:** Docker, Kubernetes, Jenkins, Ansible, Terraform, Airflow, Concourse
- **Cloud & Infrastructure:** AWS (EC2, SSM, Secrets Manager, S3, Lambda, EKS, CloudWatch, EMR, IAM, VPC), Azure (Azure VMs, Blob Storage, Azure Functions, Azure DevOps), CloudFoundry
- **Messaging & Streaming:** Apache Kafka, AWS Kinesis, AWS SQS, Azure Event Hubs
- **Databases:** MongoDB, Cassandra, MySQL, Oracle, Elasticsearch, DataStax Graph, Redshift, Cosmos DB, Dynamodb
- **Version Control & Tools:** Git, Linux/Unix, Maven, Gradle, JIRA
- **Monitoring & Observability:** Prometheus, Grafana, ELK, Newrelic, AppDynamics

## PROFESSIONAL EXPERIENCE

---

### Bank of New York Mellon

Chennai, TN, India

Lead Full Stack Developer

September 2022 - January 2025

- At Bank of New York Mellon, I led the development of client-facing portals for ticketing, creating automated workflows, and operations reporting, reducing manual support effort by 40% and response times by 35%
- I built scalable frontends in ReactJS and backends in Flask, Spring Boot, and Express, handling over 10K monthly user interactions
- Led the development of an automated kickoff framework for end-to-end infrastructure deployments using Ansible, Terraform, and Bash, enabling seamless product version migrations and dynamic environment scaling. Integrated the workflow into client-facing self-service portals, allowing on-demand provisioning with real-time observability for both clients and support teams cutting release cycles by 60% and errors by 80%
- Designed and implemented a real-time infrastructure metrics warehouse to monitor cloud and on-prem environments by aggregating metrics from deployed application servers, databases, and infrastructure components. The system collected detailed data including IP addresses, CPU/memory/disk utilization, request throughput, and uptime metrics across environments. Integrated with observability tools to power live dashboards and alerts, enabling teams to perform faster root-cause analysis (reduced by 50%), proactively manage capacity, and ensure compliance across systems.
- Additionally, I modernized legacy services by containerizing them into Docker clusters on a custom AppEngine platform, increasing service uptime by 45% and enabling faster, more reliable deployments.

### Elsevier

Chennai, TN, India

Senior Software Engineer

July 2020 - September 2022

- Architected and implemented modern search data ingestion pipelines by aggregating data from Kafka, SQS, Kinesis, document stores, S3, and REST APIs into Elasticsearch, supporting ingestion of over 10 million documents daily. Fine-tuned index mappings, analyzers, and tokenizers to deliver low-latency, high-precision search across flagship platforms like Scopus and Mendeley, improving query accuracy by 30% and reducing average search response time by 40%.
- Built a personalized reviewer recommendation system to identify domain experts for submitted research papers by analyzing author profiles, publication history, and topic similarity, enhancing the peer-review assignment process and reducing manual effort for editorial teams.

- Developed a configuration-driven semantic search API powered by entity recognition models and Elasticsearch, enabling intelligent, context-aware retrieval from structured and unstructured data sources. This reduced “no result” queries by 45% and improved user relevance satisfaction by 35%. Additionally, contributed to the DevOps lifecycle by building CI/CD pipelines with Jenkins and Ansible, deploying services to EKS (Kubernetes), and provisioning infrastructure via Terraform, cutting deployment time by 60% and minimizing manual intervention.

## **Accenture**

*Senior Software Engineer*

**Chennai, TN, India**

*August 2019 - July 2020*

- Worked as Full stack developer (Springboot-React) to develop modern microservices/micro frontends to find and resolve network interruptions. I was involved in the design, development of the applications, development of CI/CD pipelines, writing automated tests, telemetry, configuration of APM agents, collecting metrics, altering, etc.
- Designed and implemented microservices architecture using the full Spring Cloud and Spring Boot tech stack, including Eureka Service Registry, Zuul/Spring Cloud Gateway for API routing, Spring Cloud Config Server and Vault for centralized configuration and secrets management, Spring Cloud Function for serverless logic, Hystrix for circuit breaking, and Spring Security for robust authentication and authorization. Leveraged Spring for Apache Kafka for distributed messaging and Spring Data for seamless data persistence across NoSQL and relational databases.
- Migrated an existing solution written in Java/Spring to Python/Flask and realized how quick the development using Python can be. Developed a data model using Cassandra as the database since the service required very fast reads.

## **Cognizant**

*Associate*

**Chennai, TN, India**

*March 2018 - August 2019*

- Being a business product specialist (Guidewire InsuranceNow), I worked in developing solutions over the existing product to create an insurance suite, which can accept submissions, create quotes for the policy, issue the policy and also support post insurance transactions across multiple lines of business.
- Here, I developed both the front end in jQuery and Angular and the backend service that was backed by a custom homegrown servlet just like Spring's dispatcher servlet. It had all the session management, cookies, caching, validation, restful service, etc., implemented in-house.

## **Tata Consultancy Services**

*System Engineer*

**Chennai, TN, India**

*May 2015 - March 2018*

- Developing REST APIs replacing existing SOAP APIs. The service was a monolith backed by Oracle and Elasticsearch data stores.
- Enhanced the existing workspace component by integrating distributed in-memory caching using Terracotta, significantly improving data access speed and scalability. Optimized complex data management and traversal by applying appropriate data structures and algorithms. Designed and maintained Spring Batch jobs for large-scale data processing, orchestrated using Quartz Scheduler and integrated via Spring Integration to enable event-driven, distributed execution across systems.

## **EDUCATION**

---

### **San Jose State University**

*Master's, Applied Data Intelligence*

**January 2025 - December 2026**

- Pursuing education in data intelligence and analytics, inspired by rapid advancements in AI and machine learning, with a focus on building intelligent systems that solve complex, real-world problems

### **SASTRA University**

*Bachelor's, Mechanical Engineering*

**June 2011 - April 2015**

- Gained foundational engineering skills while developing a strong interest in software engineering during the final year, sparked by the challenge of solving real-world problems through programming.